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|---|------|---|----------------------|-----|
| 7 | (a) | induced e.m.f./current produces effects / acts in such a direction / tends to oppose the change causing it | M1
A1 | [2] |
| | (b) | (i) | | |
| | | 1. to reduce flux losses/increase flux linkage/easily magnetised <u>and</u> demagnetised | B1 | [1] |
| | | 2. to <u>reduce</u> energy / heat losses (<i>do not allow 'to prevent energy losses'</i>) caused by eddy currents
(<i>allow 1 mark for 'reduce eddy currents'</i>) | M1
A1 | [2] |
| | (ii) | alternating current / voltage
gives rise to (changing) flux in core
flux links the <u>secondary coil</u>
(by Faraday's law) changing flux induces e.m.f. (in secondary coil) | B1
B1
M1
A1 | [4] |
| 8 | (a) | discrete quantity / packet / quantum of energy of electromagnetic radiation | B1 | |